

# Milena Šošić

## Curriculum Vitae



### RESEARCH INTERESTS

Natural Language Processing; Large Language Models; Emotion and Sentiment Analysis; Moral Language Analysis; Language Resources for Serbian; Semantic Web and Knowledge Graphs; Computational Social Science.

### EDUCATION

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| <b>Ph.D. in Computer Science (GPA 10/10)</b>                                                              | 2020–2025        |
| University of Belgrade, Faculty of Mathematics                                                            | Belgrade, Serbia |
| Dissertation: <i>Modeling Moral and Emotional Language Aspects in Conversational Texts Classification</i> |                  |
| <b>M.Phil. in Computer Science (GPA 10/10)</b>                                                            | 2004–2010        |
| University of Belgrade, Faculty of Mathematics                                                            | Belgrade, Serbia |
| Thesis: <i>Applying classification methods on N-gram genome analysis</i>                                  |                  |
| <b>Diploma in Computer Science (GPA 8.86/10)</b>                                                          | 1997–2004        |
| University of Belgrade, Faculty of Mathematics                                                            | Belgrade, Serbia |

### PUBLICATIONS

#### ARTICLES IN INTERNATIONAL JOURNALS:

- [M22] **M. Šošić**, J. Graovac and R. Stanković. “Building an Emotion Lexicon for Serbian Using Curated Language Resources.” *Language Resources and Evaluation, Springer, 60(1), 9, 2026*. DOI
- [M22] **M. Šošić** and J. Graovac. “Effective Methods for Email Classification: Is it a Business or Personal Email?” *Computer Science and Information Systems, 19(3), 1155-1175, 2022*. DOI

#### ARTICLES IN NATIONAL JOURNALS:

- [M53] **M. Šošić**. “SRPOL – A Lexicon Based Framework for Sentiment Strength of Serbian Texts.” *Review of the NCD, 41(1), 2022*.

#### ARTICLES IN CONFERENCE PROCEEDINGS:

- [M14] R. Stanković and **M. Šošić**. “Lexicon-Based Exploration of Emotion Dynamics in TEI Novels: Case Study on *SrpELTeC* Corpus.” *Proceedings of the International Conference of the South Slavic Languages in the Digital Environment – JuDig: Thematic Collection of Papers, 2026 (Forthcoming)*.
- [M14] **M. Šošić** and R. Stanković. “Hierarchical Recognition of Moral Foundations in Serbian Literary Corpus *SrpELTeC*.” *Proceedings of the International Conference of the South Slavic Languages in the Digital Environment – JuDig: Thematic Collection of Papers, 2026 (Forthcoming)*.

#### ARTICLES IN CONFERENCE BOOKS OF ABSTRACTS:

- [M34] **M. Šošić** and R. Stanković. “Exploring Narrative Frames via Emotional and Moral Signals in Text: A Knowledge Graph Approach to Analyze Social Discourse.” *Proceedings of the Artificial Intelligence Conference, 2026, (Forthcoming)*.
- [M34] **M. Šošić**, R. Stanković and J. Graovac. “Developing Language Resources for Recognizing Moral Aspects in the Serbian Language: Multi-label Categorization through the Perspective of Moral Foundations Theory.” *Proceedings of the Artificial Intelligence Conference, 149–150, 2025*.
- [M34] **M. Šošić**, R. Stanković and J. Graovac. “Social-Emo.SR: Emotional Multi-Label Categorization of Conversational Messages from Social Networks X and Reddit.” *Proceedings of the International Conference of the South Slavic Languages in the Digital Environment – JuDig, 58–59, 2024*.
- [M34] **M. Šošić** and J. Graovac. “Svrstavanje srpskih reči u emocionalne i moralne kategorije.” *Proceedings of the Artificial Intelligence Conference, 129–130, 2023*.

### PREPRINTS AND WORK IN PROGRESS

- [WP1] **M. Šošić**, S. Petalinkar and R. Stanković. “Characterizing Emotions in Serbian Social Media: A Novel Conversational Corpus for Affective Modeling.” *(Forthcoming)*.

LANGUAGE RESOURCES

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>EmoLex.SR</b><br>Lexicon of emotionally charged words in Serbian language, annotated according to Plutchik’s model of basic emotions. <a href="#">🔗</a>                                  | Published      |
| <b>SRPOL – SentiWords.SR</b><br>Serbian sentiment intensity lexicon and related SRPOL tool for sentiment intensity estimation in textual sequences written in Serbian. <a href="#">🔗</a>    | Published      |
| <b>Social-Emo.SR</b><br>Emotion corpus of 34.6k social media messages in Serbian collected from Twitter and Reddit platforms annotated according to the Plutchik’s model of basic emotions. | In publication |
| <b>Emo-TeslaXLM</b><br>BERT-based model of the XLM architecture fine-tuned on Serbian language and emotional messages corpora with the emotion classification objective.                    | In publication |
| <b>Social-Mor.SR</b><br>Social media corpus of 13.6k messages in Serbian annotated into moral sentiment categories according to the Moral Foundations Theory (MFT).                         | In preparation |
| <b>Moral-TeslaXLM</b><br>BERT-based model of the XLM architecture fine-tuned on Serbian language and emotional messages corpora with the moral foundation classification objective.         | In preparation |
| <b>MFD.SR</b><br>Lexical resource for computational analysis of moral language in Serbian.                                                                                                  | In preparation |

CONFERENCE PRESENTATIONS

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Artificial Intelligence Conference</b><br>Serbian Academy of Sciences and Arts                                                                            | 2023, 2025<br>Belgrade, Serbia |
| <b>International Conference South Slavic Languages in the Digital Environment – JuDig</b><br>Faculty of Philology/Mining and Geology, University of Belgrade | 2024, 2026<br>Belgrade, Serbia |

RESEARCH PROJECTS

|                                                                                                          |           |
|----------------------------------------------------------------------------------------------------------|-----------|
| <b>TESLA – Text Embeddings Serbian Language Applications</b><br>Emotion Recognition for Serbian Language | 2022–2026 |
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AWARDS AND HONORS

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| <b>Annual Award in Computing for Ph.D. Students</b><br>Mathematical Institute of the Serbian Academy of Sciences and Arts | 2026 |
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PROFESSIONAL SERVICE

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>Member of multidisciplinary academic association JeRTeh</b><br>JeRTeh association is dedicated to development of language resources and technologies for Serbian language | 2022–present<br>Belgrade, Serbia |
| <b>Advisory board member</b><br>Serbian AI Society                                                                                                                           | 2020–2022<br>Belgrade, Serbia    |

PROFESSIONAL EXPERIENCE

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>Senior Data Scientist</b><br>Technical lead for advanced AI and NLP solutions, including LLM-based systems, Retrieval-Augmented Generation (RAG) pipelines, agentic AI architectures, and knowledge graph solutions for enterprise applications. | Mar, 2024 –present<br>Publicis Sapient |
| <b>Senior Data Scientist</b><br>Developed production-ready AI solutions with emphasis on NLP, predictive analytics, and applied research. Mentored junior data scientists and contributed to technical recruitment.                                 | Jan, 2021 –Mar, 2024<br>Grid Dynamics  |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>AI Team Lead</b>                                                                                                                                                                                     | Feb, 2019 –Jan, 2021    |
| Led the AI team and coordinated the development of intelligent software solutions. Introduced machine learning components into commercial products, with emphasis on text mining and sequence analysis. | Devtech                 |
| <b>Machine Learning Team Lead</b>                                                                                                                                                                       | Jan, 2016 –Feb, 2019    |
| Led the development of machine learning solutions for Product Lifecycle Management (PLM), focusing on text classification, document processing, and business intelligence applications.                 | Selerant                |
| <b>Data Scientist</b>                                                                                                                                                                                   | Dec, 2013 –Jan, 2016    |
| Developed predictive analytics models for customer segmentation, churn prediction, marketing, and risk assessment in the telecommunications domain.                                                     | Telenor                 |
| <b>Solutions Analyst</b>                                                                                                                                                                                | Dec, 2009 –Dec, 2013    |
| Designed and optimized enterprise software solutions and business processes within SAP systems, supporting global business operations.                                                                  | PepsiCo                 |
| <b>Senior Software Developer (.NET)</b>                                                                                                                                                                 | Jan, 2007 –Dec, 2009    |
| Developed and maintained software systems for electronic voting using Microsoft .NET technologies.                                                                                                      | Dominion Voting Systems |
| <b>Software Developer (.NET)</b>                                                                                                                                                                        | Aug, 2004 –Jan, 2007    |
| Developed enterprise software solutions in C# and .NET, with emphasis on RFID technologies and industrial applications.                                                                                 | SDD ITG                 |

### TECHNICAL SKILLS

**Programming:** Python, C#, SQL

**Architectures:** RAG, GraphRAG, Agentic Frameworks

**Databases:** MS SQL Server, Oracle, IBM DB2, PostgreSQL, MongoDB, Redis, FAISS, Milvus, Azure Search Index, Vertex AI Search

**Machine Learning and NLP:** PyTorch, TensorFlow, HuggingFace Transformers, spaCy, nltk

**LLMs and Tools:** GPT, Llama, Qwen, Gemma, Unsloth, Promptflow, LangChain, LangGraph, DSPy, LangFuse, Ragas, DeepEval

**Knowledge Graphs:** RDF, OWL, SPARQL, Neo4j, Cypher

**MLOps:** Docker, Git, GitHub Actions, vLLM, LiteLLM

**Cloud Platforms:** Azure, GCP, AWS

**Data and Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly

**Typesetting:**  $\LaTeX$

### LANGUAGES

**Serbian:** Native

**English:** Full professional proficiency

### PERSONAL SKILLS AND INTERESTS

**Characteristics:** Organized, detail-oriented, self-motivated and collaborative.

**Hobbies:** Hiking, reading, playing flute, traveling and exploring new technologies.